SCENARIO-BASED TEST RESULTS

Purdue-Based Injection Substation OT/SCADA Monitoring Demonstrator

1. Project Context

This report records observations from the Purdue-based injection substation OT/SCADA monitoring demonstrator developed in Ignition Designer using Perspective. The six scenarios are selected through a scenario tag and represented on the architecture dashboard. These results describe visible behaviour in the demonstrator; they do not establish detection of real industrial network failures or validate live cybersecurity controls.

2. Test Results

ID	Scenario	Affected asset shown	Impact shown	Observed dashboard behaviour	Result
0	Normal Operation	None	No Impact	Displayed indicators are green; devices show online; Substation LAN is healthy.	Observed — baseline
1	Substation LAN Failure	Substation LAN	Loss of Operational Communications	Substation LAN shows FAILED and PLC/RTU shows OFFLINE.	Observed
2	Historian Failure	Historian	Loss of Historical Data Recording	OT Historian shows OFFLINE while Substation LAN remains healthy.	Observed
3	OT Firewall Failure	OT Firewall	OT Boundary Degraded	OT Firewall shows OFFLINE; IT Firewall, PLC/RTU, Bay Control IED, OT Historian, and Substation LAN remain shown as online/healthy.	Observed — consistent with intended design
4	Enterprise Firewall Failure	Enterprise Firewall	IT/OT Boundary Degraded	IT Firewall shows OFFLINE; OT Firewall and Bay Control IED remain ONLINE after correction.	Observed — corrected behaviour
5	IED Failure	Protection IED	Reduced Protection Visibility	Protection IED shows OFFLINE. The GOOSE label still shows HEALTHY and is currently static.	Observed — GOOSE label is not live status

3. Findings

• The scenario selector updates the displayed scenario, affected asset, and impact description.

• The dashboard uses visual status indicators to represent scenario-dependent asset and network states.

• Scenario 3 represents an OT Firewall failure without automatically marking the Bay Control IED offline.

• Scenario 4 was corrected so the Bay Control IED remains online during Enterprise Firewall Failure.

• The IEC 61850 GOOSE status currently appears as a static architecture label. It should not be described as a live communication-health measurement.

4. Scope and Limitations

These results are based on the displayed state of the Ignition demonstrator. They do not prove that real network failures are detected, that the named industrial protocols are exchanging live traffic, or that cybersecurity controls have been validated against actual attacks. The architecture includes protocol labels and design concepts. Actual protocol integration, packet capture and analysis, and controlled cybersecurity experiments should be documented separately only after implementation and testing.

5. Recommended Follow-Up Validation

1. Confirm that each scenario continues to produce the intended displayed state after changing away from it and selecting it again.

2. Verify the tag or expression dependencies for the affected status indicators.

3. Decide whether GOOSE should remain a static architecture annotation, become a clearly labelled simulated status, or eventually be supported by protocol-level observations in an isolated lab.

4. Record unexpected results and corrective actions with screenshots.

5. Update this report when further testing is completed.

6. Evidence

Store screenshots for Scenarios 0–5 in a dedicated repository folder, for example `screenshots/scenario-tests/`, and use filenames that identify each scenario. Keep the filenames consistent with any image links added to this report.